

Networks Lab Assignment 2 Report

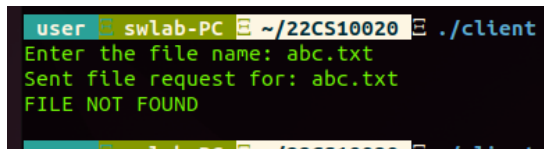
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1. Client and Server Terminals

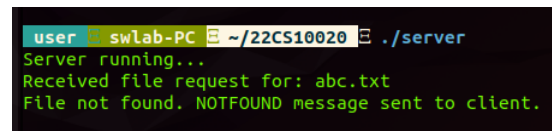
Scenario 1: File Not Available in the Local Directory

Below are the terminal screenshots for the client and server when the file was not available in the local directory:



```
user@swlab-PC: ~/22CS10020
./client
Enter the file name: abc.txt
Sent file request for: abc.txt
FILE NOT FOUND
```

(a) Client terminal output.



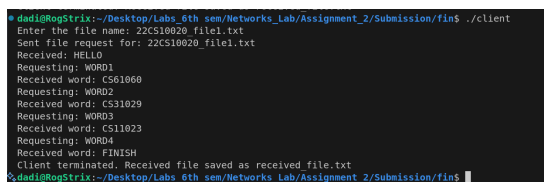
```
user@swlab-PC: ~/22CS10020
./server
Server running...
Received file request for: abc.txt
File not found. NOTFOUND message sent to client.
```

(b) Server terminal output.

Figure 1: Client and server terminals when the file was not available.

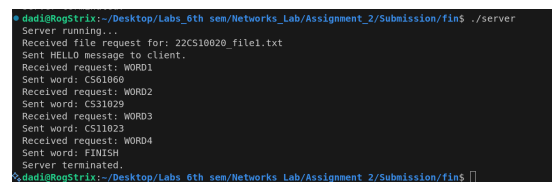
Scenario 2: File Available in the Local Directory

Below are the terminal screenshots for the client and server when the file was available in the local directory:



```
dadi@RagStrix: ~/Desktop/Labs_6th sem/Networks_Lab/Assignment_2/Submission/fin$ ./client
Enter the file name: 22CS10020_file1.txt
Sent file request for: 22CS10020_file1.txt
Received: HELLO
Requesting: WORD1
Received word: CS31060
Requesting: WORD2
Received word: CS31029
Requesting: WORD3
Received word: CS11023
Requesting: WORD4
Received word: FINISH
Client terminated. Received file saved as received_file.txt
dadi@RagStrix: ~/Desktop/Labs_6th sem/Networks_Lab/Assignment_2/Submission/fin$
```

(a) Client terminal output.



```
dadi@RagStrix: ~/Desktop/Labs_6th sem/Networks_Lab/Assignment_2/Submission/fin$ ./server
Server running...
Received file request for: 22CS10020_file1.txt
Sent HELLO message to client.
Received request: WORD1
Sent word: CS31060
Received request: WORD2
Sent word: CS31029
Received request: WORD3
Sent word: CS11023
Received request: WORD4
Sent word: FINISH
Server terminated.
dadi@RagStrix: ~/Desktop/Labs_6th sem/Networks_Lab/Assignment_2/Submission/fin$
```

(b) Server terminal output.

Figure 2: Client and server terminals when the file was available.

2. Packet Exchange Screenshots

Scenario 1: File Not Available in the Local Directory

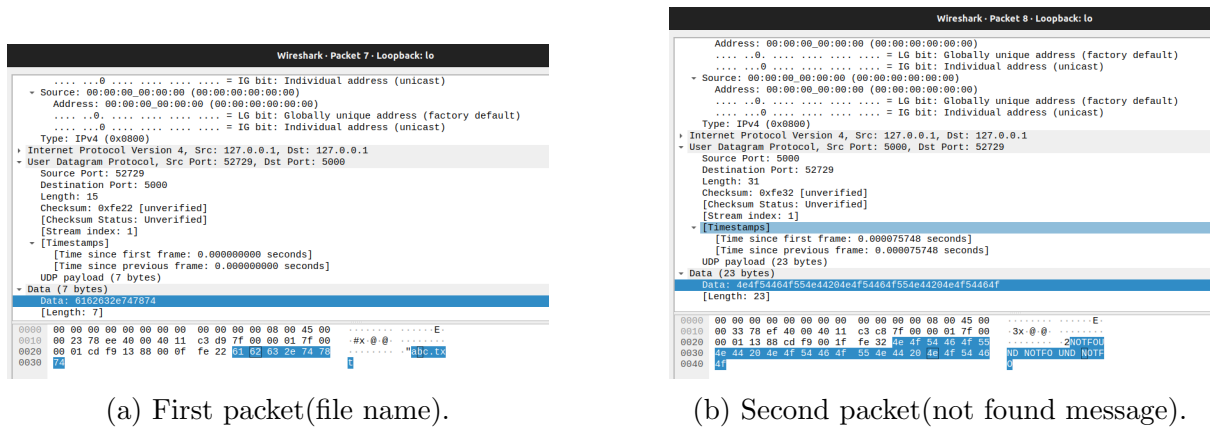


Figure 3: Packets exchanged when the file was not available.

Scenario 2: File Available in the Local Directory



Figure 4: Initial phase: File request and first word transfer

```

Wireshark - Packet 171 - Loopback: lo
+ Frame 171: 47 bytes on wire (376 bits), 47 bytes captured (376 bits) on interface lo, id 0
+ Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
+ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
+ User Datagram Protocol, Src Port: 46072, Dst Port: 5001
  Source Port: 46072
  Destination Port: 5001
  Length: 13
  Checksum: 0xfe20 [unverified]
  [Checksum Status: Unverified]
  [Stream Index: 39]
  [Stream Packet Number: 5]
  [Timestamps]
  UDP payload (5 bytes)
  Data (5 bytes)
  Data: 574f524432
  [Length: 5]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E
0010  00 21 3f bc 40 00 40 11 fd 11 7f 00 00 01 7f 00  ..? @ @ ...
0020  00 01 b3 f6 13 00 00 00 fe 20 37 4f 52 44 32  .... WORD2

```

(a) Packet 5: Client requests WORD2

```

Wireshark - Packet 173 - Loopback: lo
+ Frame 173: 47 bytes on wire (376 bits), 47 bytes captured (376 bits) on interface lo, id 0
+ Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
+ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
+ User Datagram Protocol, Src Port: 46072, Dst Port: 5001
  Source Port: 46072
  Destination Port: 5001
  Length: 13
  Checksum: 0xfe20 [unverified]
  [Checksum Status: Unverified]
  [Stream Index: 39]
  [Stream Packet Number: 7]
  [Timestamps]
  UDP payload (5 bytes)
  Data (5 bytes)
  Data: 574f524433
  [Length: 5]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E
0010  00 21 3f bc 40 00 40 11 fd 0f 7f 00 00 01 7f 00  ..? @ @ ...
0020  00 01 b3 f6 13 00 00 00 fe 20 37 4f 52 44 33  .... WORD3

```

(c) Packet 7: Client requests WORD3

```

Wireshark - Packet 175 - Loopback: lo
+ Frame 175: 47 bytes on wire (376 bits), 47 bytes captured (376 bits) on interface lo, id 0
+ Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
+ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
+ User Datagram Protocol, Src Port: 46072, Dst Port: 5001
  Source Port: 46072
  Destination Port: 5001
  Length: 13
  Checksum: 0xfe20 [unverified]
  [Checksum Status: Unverified]
  [Stream Index: 39]
  [Stream Packet Number: 9]
  [Timestamps]
  UDP payload (5 bytes)
  Data (5 bytes)
  Data: 574f524434
  [Length: 5]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E
0010  00 21 3f bc 40 00 40 11 fd 0f 7f 00 00 01 7f 00  ..? @ @ ...
0020  00 01 b3 f6 13 00 00 00 fe 20 37 4f 52 44 34  .... WORD4

```

(a) Packet 9: Client requests WORD4

```

Wireshark - Packet 172 - Loopback: lo
+ Frame 172: 49 bytes on wire (392 bits), 49 bytes captured (392 bits) on interface lo, id 0
+ Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
+ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
+ User Datagram Protocol, Src Port: 5001, Dst Port: 46072
  Source Port: 5001
  Destination Port: 46072
  Length: 15
  Checksum: 0xfe22 [unverified]
  [Checksum Status: Unverified]
  [Stream Index: 39]
  [Stream Packet Number: 6]
  [Timestamps]
  UDP payload (7 bytes)
  Data (7 bytes)
  Data: 43533331303239
  [Length: 7]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E
0010  00 22 3f bd 40 00 40 11 fd 0e 7f 00 00 01 7f 00  ..? @ @ ...
0020  00 01 13 89 b3 f6 00 0f fe 22 43 53 33 31 30 32  .... "CS3102
0030  39 9

```

(b) Packet 6: Server sends second word "CS31029"

```

Wireshark - Packet 174 - Loopback: lo
+ Frame 174: 49 bytes on wire (392 bits), 49 bytes captured (392 bits) on interface lo, id 0
+ Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
+ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
+ User Datagram Protocol, Src Port: 5001, Dst Port: 46072
  Source Port: 5001
  Destination Port: 46072
  Length: 15
  Checksum: 0xfe22 [unverified]
  [Checksum Status: Unverified]
  [Stream Index: 39]
  [Stream Packet Number: 8]
  [Timestamps]
  UDP payload (7 bytes)
  Data (7 bytes)
  Data: 43533331303233
  [Length: 7]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E
0010  00 22 3f bd 40 00 40 11 fd 0e 7f 00 00 01 7f 00  ..? @ @ ...
0020  00 01 13 89 b3 f6 00 0f fe 22 43 53 33 31 30 32  .... "CS1102
0030  33 3

```

(d) Packet 8: Server sends third word "CS11023"

```

Wireshark - Packet 176 - Loopback: lo
+ Frame 176: 48 bytes on wire (384 bits), 48 bytes captured (384 bits) on interface lo, id 0
+ Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
+ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
+ User Datagram Protocol, Src Port: 5001, Dst Port: 46072
  Source Port: 5001
  Destination Port: 46072
  Length: 14
  Checksum: 0xfe21 [unverified]
  [Checksum Status: Unverified]
  [Stream Index: 39]
  [Stream Packet Number: 10]
  [Timestamps]
  UDP payload (6 bytes)
  Data (6 bytes)
  Data: 46494e495348
  [Length: 6]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E
0010  00 22 3f bd 40 00 40 11 fd 0e 7f 00 00 01 7f 00  ..? @ @ ...
0020  00 01 13 89 b3 f6 00 0e fe 21 46 49 4e 49 53 48  .... FINISH

```

(b) Packet 10: Server sends "FINISH", ending transfer

Figure 6: Final phase: Last word request and completion

3. Protocol Used

Task: Identify the protocol used for communication.

Protocol: *UDP*

4. Source and Destination Details

Task: Identify the IP addresses and ports of the source and destination.

When the File Was Available	
Client IP and Port:	<i>127.0.0.1, PORT: 46072</i>
Server IP and Port:	<i>127.0.0.1, PORT: 5001</i>
When the File Was Not Available	
Client IP and Port:	<i>127.0.0.1, PORT: 52729</i>
Server IP and Port:	<i>127.0.0.1, PORT: 5001</i>

Table 1: Client and Server IP and Port Information

5. Packet Sizes

Task 1: Measure the size of the FILENAME request sent by the client.

Size: *19 bytes*

Task 2: Measure the size of the server's responses for HELLO and the first word (*WORD1*).

HELLO Response Size: *47 bytes on the wire (376 bits), 47 bytes (376 bits) captured on interface lo, id 0*

WORD1 Response Size: *49 bytes on the wire (376 bits), 47 bytes (376 bits) captured on interface lo, id 0*

6. Payload Inspection

Task: Inspect the payload of packets where words are transmitted.

Payload Details: The UDP payloads of packets transmitting the words are shown below. Each packet contains a specific payload, as visible in the captured data. The hexadecimal representation and its ASCII equivalent are included for each word.

```

Wireshark - Packet 168 - assignment_2.pcap
* Frame 168: 47 bytes on wire (376 bits), 47 bytes captured (376 bits)
* Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
* Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
* User Datagram Protocol, Src Port: 5001, Dst Port: 46072
  Source Port: 5001
  Destination Port: 46072
  Length: 15
  Checksum: 0xfe20 [unverified]
  [Checksum Status: Unverified]
  [Stream Index: 39]
  [Stream Packet Number: 2]
  [Timestamps]
    [Time since first frame: 0.000151000 seconds]
    [Time since previous frame: 0.000151000 seconds]
  UDP payload (5 bytes)
  Data (5 bytes)
    Data: 40454c4cf
    [Length: 5]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E.
0010  00 21 3f b5 40 00 40 11  fd 14 7f 00 00 01 7f 00  ..?@?
0020  00 01 13 89 b3 f8 00 0e  fe 20 40 45 4c 4c 4f  HELLO

```

(a) UDP payload containing the word HELLO.

```

Wireshark - Packet 170 - assignment_2.pcap
* Frame 170: 49 bytes on wire (392 bits), 49 bytes captured (392 bits)
* Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
* Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
* User Datagram Protocol, Src Port: 5001, Dst Port: 46072
  Source Port: 5001
  Destination Port: 46072
  Length: 15
  Checksum: 0xfe22 [unverified]
  [Checksum Status: Unverified]
  [Stream Index: 39]
  [Stream Packet Number: 4]
  [Timestamps]
    [Time since first frame: 0.000342000 seconds]
    [Time since previous frame: 0.000020000 seconds]
  UDP payload (7 bytes)
  Data (7 bytes)
    Data: 43533631303630
    [Length: 7]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E
0010  00 23 3f b7 40 00 40 11  fd 19 7f 00 00 01 7f 00  ..?@?
0020  00 01 13 89 b3 f8 00 0f  fe 22 43 53 36 31 30 36  ..CS6106
0030  39 0

```

(b) UDP payload containing the word CS61060.

```

Wireshark - Packet 172 - assignment_2.pcap
* Frame 172: 49 bytes on wire (392 bits), 49 bytes captured (392 bits)
* Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
* Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
* User Datagram Protocol, Src Port: 5001, Dst Port: 46072
  Source Port: 5001
  Destination Port: 46072
  Length: 15
  Checksum: 0xfe22 [unverified]
  [Checksum Status: Unverified]
  [Stream Index: 39]
  [Stream Packet Number: 6]
  [Timestamps]
    [Time since first frame: 0.000480000 seconds]
    [Time since previous frame: 0.000036000 seconds]
  UDP payload (7 bytes)
  Data (7 bytes)
    Data: 43533331303239
    [Length: 7]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E
0010  00 23 3f b5 40 00 40 11  fd 0e 7f 00 00 01 7f 00  ..?@?
0020  00 01 13 89 b3 f8 00 0f  fe 22 43 53 33 31 30 32  ..CS1029
0030  39 9

```

(a) UDP payload containing the word CS31029.

```

Wireshark - Packet 174 - assignment_2.pcap
* Frame 174: 49 bytes on wire (392 bits), 49 bytes captured (392 bits)
* Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
* Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
* User Datagram Protocol, Src Port: 5001, Dst Port: 46072
  Source Port: 5001
  Destination Port: 46072
  Length: 15
  Checksum: 0xfe22 [unverified]
  [Checksum Status: Unverified]
  [Stream Index: 39]
  [Stream Packet Number: 8]
  [Timestamps]
    [Time since first frame: 0.000500000 seconds]
    [Time since previous frame: 0.000074000 seconds]
  UDP payload (7 bytes)
  Data (7 bytes)
    Data: 43533331303233
    [Length: 7]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E
0010  00 23 3f b5 40 00 40 11  fd 0e 7f 00 00 01 7f 00  ..?@?
0020  00 01 13 89 b3 f8 00 0f  fe 22 43 53 31 31 30 32  ..CS1102
0030  33 3

```

(b) UDP payload containing the word CS11023.

```

Wireshark - Packet 176 - assignment_2.pcap
* Frame 176: 48 bytes on wire (384 bits), 48 bytes captured (384 bits)
* Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
* Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
* User Datagram Protocol, Src Port: 5001, Dst Port: 46072
  Source Port: 5001
  Destination Port: 46072
  Length: 14
  Checksum: 0xfe21 [unverified]
  [Checksum Status: Unverified]
  [Stream Index: 39]
  [Stream Packet Number: 10]
  [Timestamps]
    [Time since first frame: 0.000580000 seconds]
    [Time since previous frame: 0.000034000 seconds]
  UDP payload (6 bytes)
  Data (6 bytes)
    Data: 46494e495348
    [Length: 6]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E
0010  00 22 3f bd 40 00 40 11  fd 0b 7f 00 00 01 7f 00  ..?@?
0020  00 01 13 89 b3 f8 00 0e  fe 21 46 49 4e 49 53 48  ..FINISH

```

(a) UDP payload containing the word FINISH.

7. Total File Transfer Time

Task: Measure the total time taken for the file transfer from start to finish.

Total Time: *0.000507 seconds*

8. Average Packet Size

Task: Calculate the average size of each packet during communication.

Average Packet Size: *47.78 bytes*